# Hyperbola-Band Attention Supervision for Detection in GPR B-Scan Images

IEEE Publication Technology, *Staff, IEEE,*

***Abstract*—This document describes the most common article elements and how to use the IEEEtran class with LaTeX to produce files that are suitable for submission to the IEEE. IEEEtran can produce conference, journal, and technical note (correspondence) papers with a suitable choice of class options.**

***Index Terms*—Article submission, IEEE, IEEEtran, journal, LaTeX, paper, template, typesetting.**

## I. Introduction

With the growing demand for refined management of urban underground space, service condition assessment of transportation infrastructure, and geotechnical engineering investigation, efficient and reliable underground nondestructive testing technologies have become a prominent research direction in the engineering field [1]. Ground Penetrating Radar (GPR), favored for its non-intrusive nature, high survey efficiency, and moderate spatial resolution, is widely applied in scenarios such as underground pipeline detection, road internal defect inspection, tunnel lining quality assessment, and concealed engineering investigation [2], [3]. During field surveys, the antenna moves along a survey line, emitting electromagnetic pulses and collecting reflected echoes from subsurface media at each position. By stacking consecutive single-trace echoes side by side along the survey line, a two-dimensional B-scan image is obtained, where the horizontal axis corresponds to the spatial position of the antenna, and the vertical axis represents the two-way travel time of electromagnetic waves, which can be equivalently converted to subsurface depth.

For compact subsurface targets such as pipelines and rebars, the round-trip path of electromagnetic waves is the shortest when the antenna is directly above the target, corresponding to the minimum travel time. As the antenna moves laterally away from the target, the path length increases symmetrically and the travel time rises accordingly. As a result, the reflected echoes of such targets appear as downward-opening hyperbolas in B-scan images, with the apex corresponding to the vertical projection of the target. For this reason, hyperbola detection serves as a critical prerequisite for subsurface target localization, depth estimation, and parameter reconstruction [4].

Early hyperbola detection methods mainly relied on hand-crafted features and analytical curve fitting, typical examples including Hough transform voting, predefined template matching, and HOG features combined with SVM classification [3]. While these methods offer clear interpretability under ideal experimental conditions, they suffer from notable practical limitations: they incur high computational costs, are sensitive to the discretization precision of parameters, and their performance heavily depends on manually designed prior templates [4]. When applied to field data with heterogeneous subsurface media and strong clutter interference, their performance degrades significantly.

With the development of deep learning, end-to-end object detection methods have been widely adapted for B-scan hyperbola recognition [5]–[8], delivering remarkable improvements in detection efficiency and localization accuracy. However, existing studies generally adopt rectangular bounding boxes designed for natural images as the annotation and output format [6], without tailoring the scheme to the geometric properties of hyperbolas. A rectangular box can only roughly delineate the spatial range of an echo, enclosing a large amount of non-target background area. It neither provides fine-grained geometric supervision such as vertex position, curvature, or band thickness, nor enables the model to focus on learning the structural characteristics of hyperbolas themselves, which restricts further improvement of detection accuracy and leads to poor model stability in scenarios with limited data.

Meanwhile, attention mechanisms have become a common approach to enhancing GPR hyperbola detection performance [9]–[14]. They reweight features along the channel or spatial dimensions to amplify target-relevant features and suppress irrelevant background noise. Nevertheless, due to the lack of explicit guidance from target geometric priors, their performance remains inconsistent across different scenarios.

To address the aforementioned limitations, this paper proposes a hyperbola-band attention supervision method. On the annotation side, the geometric shape of each hyperbola is encoded with a small set of parameters; on

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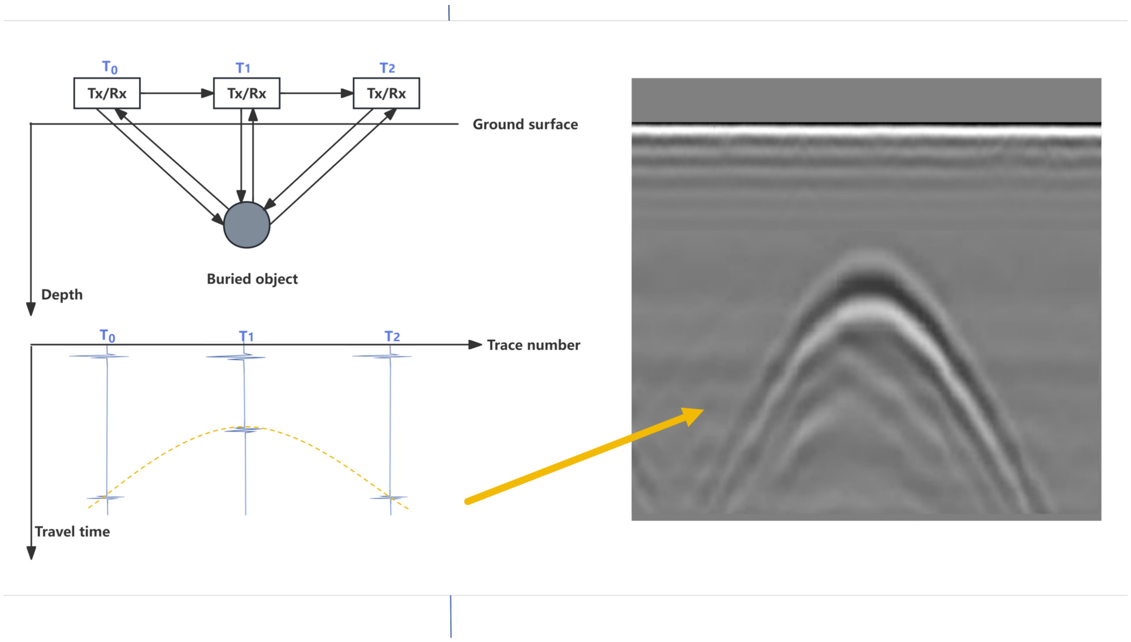



Fig. 1. Formation of a GPR B-scan.

the learning side, this geometric information is converted into supervisory signals for an attention branch to boost detection performance.

The main contributions of this work are summarized as follows:

1) A dedicated annotation tool is developed for hyperbola-band targets. It defines each hyperbolic band using a small set of interpretable parameters, including vertex coordinates, slope, horizontal span, and band thickness, supports interactive adjustment, and exports both shape parameters and derived rectangular bounding boxes simultaneously.
2) An explicit attention supervision mechanism based on geometric priors is proposed. Hyperbola-band masks generated from parametric annotations directly constrain the attention distribution, injecting explicit spatial priors of targets into the detection network and improving detection performance via end-to-end training.
3) Experiments on a public GPR dataset show that the proposed method improves the F1-score from 0.68 to 0.84 at a training ratio of 0.7, with a notable enhancement in model stability. It also outperforms multiple mainstream attention modules, which validates the effectiveness of the proposed method.

## II. Related Work

### A. Hyperbola recognition task

Ground-penetrating radar (GPR) forms a subsurface image by moving a transmitter/receiver (Tx/Rx) antenna along a survey line: at each position $T_0, T_1, T_2, \ldots$ it emits an electromagnetic pulse and records the reflected signal as a single trace (an A-scan) of amplitude versus two-way travel time. Stacking consecutive traces side by side yields a B-scan, whose horizontal axis is the trace number (antenna position) and whose vertical axis is travel time, which is monotonically related to depth. Because a compact buried target is illuminated from many antenna positions, its round-trip distance is shortest directly above the object and increases symmetrically to either side, so the target's reflection is imaged as a downward-opening hyperbola (Fig. 1).

In GPR B-scan images, buried pipes, cables, and roots typically appear as hyperbolic reflections, making hyperbola detection a prerequisite for target localization, depth estimation, and parameter reconstruction [15]–[17]. Early methods relied on hand-crafted features and analytical fitting, screening candidates with neural preprocessing, Viola–Jones, or HOG+SVM and then localizing them via the Hough transform or template/least-squares fitting [17]–[21]. Such pipelines work in constrained settings but remain computationally expensive, sensitive to parameter discretization, and dependent on prior template design or user intervention [16], [20], [22], [23]; later refinements combined edge detection and

C3 clustering with Hough fitting to improve robustness [23], [24].

Deep learning shifted the field from sliding-window recognition toward end-to-end detection on full B-scans. Two-stage detectors led this transition: Faster R-CNN, trained with simulated radargrams to offset scarce field labels [22], was later coupled with explicit hyperbola fitting for automatic localization [25], while a CNN-LSTM reformulated diameter identification as region classification and reached 92.5% accuracy on field data [16]. One-stage detectors emphasized real-time deployment, with YOLOv3 running at about 12 fps [26], combined with migration-based focusing and apex extraction to estimate pipeline depth within 0.04 m [27], and a YOLOv4 variant tailored to root recognition and localization [28].

More recent work argues that a generic axis-aligned box is suboptimal for a structured geometric echo, a limitation inherent to mainstream detectors that regress box bounds or centers [29], [30]. GPR-specific methods therefore inject geometric priors: anchor-optimized Mask Scoring R-CNN [31], GPR-Former, which directly regresses hyperbola parameters and refines them with a symmetry constraint [15], and shape-aware topological representations fused with YOLOv5 [32]. Overall, the field has progressed from hand-crafted curve search to geometry-aware detection, yet most detectors still localize hyperbolas through rectangular proxies rather than modeling their strip-like, symmetric, parametric shape explicitly.

### B. Attention Mechanisms in GPR Hyperbola Interpretation

Attention mechanisms are increasingly embedded into task-specific GPR pipelines so that models respond to hyperbola-relevant structures rather than processing the whole radargram uniformly. Localized hyperbolic attention enhances hyperbolic wave features and suppresses clutter to improve downstream imaging of target pipelines [33], while a dual-cascade network first removes direct-wave interference and then applies attention with feature fusion to raise hyperbola detection accuracy before geometric localization [34]. Because hyperbolic reflections extend across neighbouring traces and the full time window, transformer-style self-attention is also used to model the long-range dependencies that convolution captures less efficiently: GPR-Former regresses hyperbola parameters directly from B-scans and refines them with a symmetry constraint [15], and GPR-TransUNet introduces self-attention into a TransUNet-style regression framework for the same reason [35].

More recent designs tailor attention specifically to hyperbolic geometry and apex-localization objectives that are closely aligned with our study. A rebar-recognition model emphasizes hyperbolic features with deformable attention and a hyperbolic attention transformer before proposal generation [36], a lightweight dual-task keypoint detector couples attention-enhanced features with keypoint regression for precise vertex localization [37], and a state-space detector for arbitrarily long B-scans targets hyperbola vertex estimation directly [38]. The clear trend is a shift from generic channel or spatial reweighting toward physically informed designs that encode hyperbolic structure and apex localization, while traditional fitting-based pipelines remain relevant for their geometric interpretability [25]. This motivates our approach: instead of localizing the echo through rectangular proxies or post-hoc curve fitting, we represent the hyperbola by its strip-like, symmetric, parametric shape, so that both annotation and learning are aligned with the geometry of the target.

### C. Labelling tools

Supervised detectors depend on large sets of manually annotated images, and a range of general-purpose annotation tools has been developed to meet this need. Early web-based systems such as LabelMe established collaborative image labelling at scale [39], and the annotation pipeline behind large benchmarks like MS-COCO further standardized bounding-box, polygon, and segmentation-mask labelling [40]. Building on these efforts, widely used open-source tools including LabelImg [41], Labelme [42], the VGG Image Annotator [43], and CVAT [44] provide interactive interfaces for the common geometric primitives—axis-aligned rectangles, polygons, key points, and pixel-wise masks. These primitives are generic and domain-agnostic, which is exactly what makes the tools broadly applicable across natural-image tasks.

In GPR interpretation, hyperbolic reflections are usually annotated with these same general-purpose tools; bounding boxes drawn in software such as LabelImg are the de facto standard for labelling subsurface targets in B-scans [45]. A rectangle, however, records only the extent of the echo rather than its strip-like, symmetric, and parametric geometry, so the resulting labels are misaligned with the physical shape of the hyperbola and cannot directly provide geometric supervision such as vertex position, curvature, or band thickness. Polygon and mask tools can trace the curve more tightly, but they require dense per-instance clicking and still do not expose the underlying hyperbola parameters. This gap between the geometry-agnostic primitives offered by existing tools and the parametric, band-shaped nature of hyperbolic targets motivates the dedicated annotation tool proposed in this work, which lets annotators place and adjust a parametric hyperbola band through a few interpretable controls and export both the shape parameters and the derived bounding box.

## III. Proposed method

### A. Hyperbola annotation

We designed a visual annotation tool for hyperbolic ribbon-shaped targets, whose user interface is illustrated in the Fig. 2. First, the user selects an image folder, and the tool automatically loads all images within it. The original image is displayed on the lower-left panel, while the annotated image is shown on the right. A mouse click on the original image marks the vertex position of the hyperbola, and five adjustable parameters are provided on the right panel.

The five parameters are defined as follows: the horizontal coordinate $x_{\text{vertex}}$ and vertical coordinate $y_{\text{vertex}}$ of the hyperbola vertex, the slope $s$, the horizontal span, and the ribbon thickness. Among these, the slope $s$ is the only parameter that governs the shape of the curve. Once the vertex is placed, a single slope value fully fixes the openness of the two arms, so a larger slope yields a narrower and sharper hyperbola and a smaller slope yields a flatter one. The centreline of the downward-opening band is

$$y_c(x) = y_{\text{vertex}} \sqrt{1 + \left( \frac{s\,(x - x_{\text{vertex}})}{y_{\text{vertex}}} \right)^2}, \qquad (1)$$

and the band is the vertical ribbon of all points $(x, y)$ satisfying $|y - y_c(x)| \leq \text{thickness}/2$, where the span sets the horizontal extent over which the two arms are drawn. By clicking the *Add as New Annotation* button, the current parameter configuration is saved as a new annotation for the active image, and multiple annotations can be added to a single image.

All annotations are stored in JSON format with the following structure:

```
{
  "line2888_0-100_jpg.rf.ce38dc3799
  2454823c8dea9e0f814ac5.jpg": [
    {
      "label": "hyperbola",
      "x_vertex": 315.4,
      "y_vertex": 241.5,
      "slope": 3.09,
      "span": 387.9,
      "thickness": 115.8
    }]
}
```

Here, the top-level key denotes the image filename, and each list entry stores one annotated hyperbola band as its parameters and their corresponding values.

After the hyperbola band is annotated, the conventional bounding-box annotation can be directly derived from the boundary of the band. Fig. 3 shows an example, presenting the raw image, the annotated hyperbola band, and the corresponding bounding-box form.

### B. Hyperbola-attention nerual network

The network takes a single-channel grayscale B-scan image as input and outputs a rectangular bounding box for each hyperbolic signature in the image. Unlike existing methods, we introduce within the backbone an attention branch that is explicitly supervised by a hyperbola-band mask, and inject it into the bottleneck features via channel-wise concatenation, thereby guiding the network to focus on the hyperbolas and improving detection performance. A schematic of the network is shown in Fig. 4.

The backbone consists of three down-sampling stages followed by a bottleneck layer; each stage comprises two $3\times3$ convolutions, each followed by batch normalization and ReLU, together with a subsequent $2\times2$ max-pooling operation that doubles the channel depth while halving the spatial resolution. Starting from the $1 \times 640^2$ input, the three stages produce feature maps $f_1$ of size $32 \times 320^2$, $f_2$ of size $64 \times 160^2$, and $f_3$ of size $128 \times 160^2$, which are further reduced to the bottleneck feature $F_b$ of size $256\times80^2$. The attention branch takes $f_3$ as input rather than the bottleneck feature $F_b$, as $f_3$ retains a higher spatial resolution that is better suited to precise localization.

The supervised attention branch is the central component of the proposed method. A lightweight head $g_a(\cdot)$, consisting of a $3 \times 3$ convolution with batch normalization and ReLU followed by a $1\times1$ convolution, maps $f_3$ to a single-channel logit that is passed through a sigmoid to yield the attention map $A \in [0, 1]^{H\times W}$, whose entry $A_i$ is the predicted probability that pixel $i$ belongs to the hyperbola band. The supervisory target $B \in \{0, 1\}^{H\times W}$ is a binary band mask rasterized from the annotated hyperbola geometry at the same resolution. The attention map $A$ is then driven to approximate $B$ by a loss that combines a binary cross-entropy term and a Dice term, both computed between $A$ and $B$ over the $N = HW$ pixels,

$$\mathcal{L}_{\text{att}} = \mathcal{L}_{\text{bce}} + \mathcal{L}_{\text{dice}}, \qquad (2)$$

$$\mathcal{L}_{\text{bce}} = -\frac{1}{N} \sum_{i=1}^{N} \Big[ B_i \log A_i + (1-B_i) \log(1-A_i) \Big], \qquad (3)$$

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2 \sum_{i=1}^{N} A_i B_i}{\sum_{i=1}^{N} A_i + \sum_{i=1}^{N} B_i + \epsilon}, \qquad (4)$$

where $\epsilon = 10^{-6}$ is a small constant added to the denominator to avoid division by zero. The cross-entropy term performs an independent per-pixel classification, penalizing each pixel that is misclassified as belonging

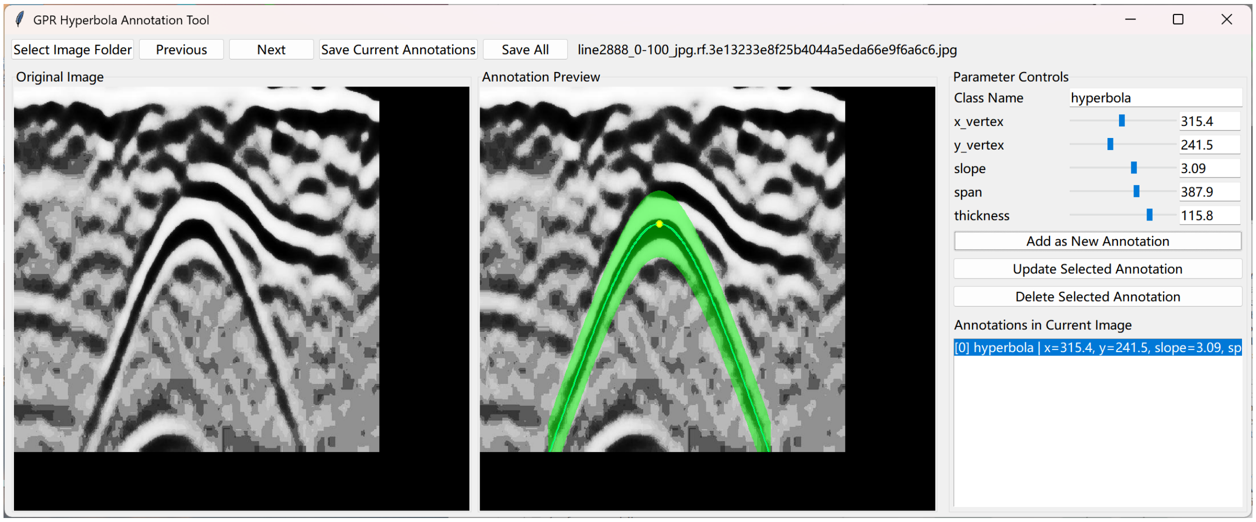
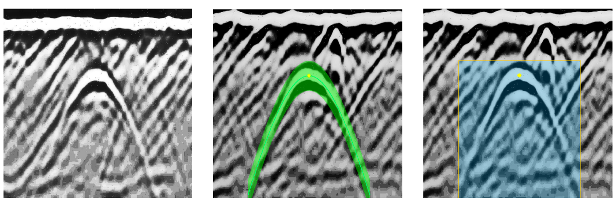



Fig. 2. tool



(a) Raw image. (b) Annotated hyperbola band. (c) Derived bounding box.

Fig. 3. Example of a hyperbola sample: the raw image, the annotated hyperbola band, and the bounding box derived from the band boundary.

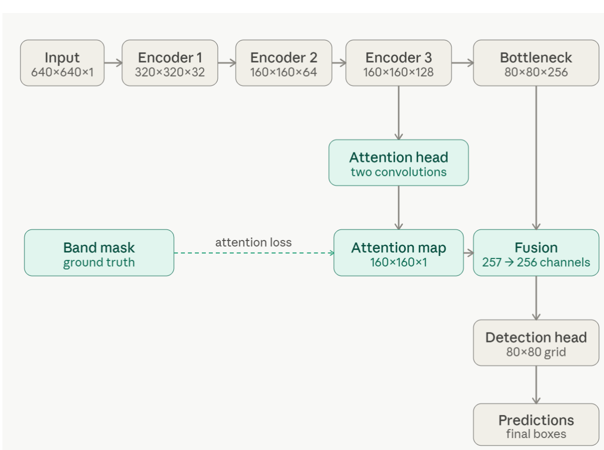



Fig. 4. 11

to, or not belonging to, the hyperbola band. The Dice term instead measures the overall overlap between $A$ and $B$, with the loss decreasing as the overlap increases. The two terms are complementary: the former enforces pixel-level accuracy, while the latter enforces region-level shape consistency and helps mitigate the class imbalance caused by the band occupying only a small fraction of the image.

*a) Attention-guided feature fusion.:* The attention map $A$ is passed through a sigmoid and downsampled by a factor of two via average pooling to align with the bottleneck resolution, yielding $\tilde{A}$. It is then concatenated with $F_b$ along the channel dimension and compressed back to 256 channels by a $1 \times 1$ convolution with batch normalization and ReLU, producing the fused feature $\hat{F}$. This operation injects the spatial prior of the hyperbola band into the feature representation passed to the detection head.

*b) Detection head and decoding.:* The detection head follows a YOLO-style dense prediction paradigm, producing three parallel outputs over an $80 \times 80$ grid at stride 8: objectness (1 channel), box size (2 channels), and sub-pixel center offset (2 channels). During inference, grid cells whose sigmoid-activated objectness exceeds a threshold $\tau$ are decoded into bounding boxes, and redundant detections are removed by box-level IoU-based non-maximum suppression. Let $\mathcal{P}$ denote the set of positive cells, namely the grid cells that contain an object center, and $\mathcal{N}$ the remaining background cells. The objectness map is supervised by a class-balanced binary cross-entropy over the hard targets $p_j$, which equal 1 at positive cells and 0 elsewhere,

$$\mathcal{L}_{\mathrm{obj}} = \frac{1}{|\mathcal{P}|}\sum_{j\in\mathcal{P}} \mathrm{BCE}(\hat{o}_j, 1) + \frac{\lambda_{\mathrm{noobj}}}{|\mathcal{N}|}\sum_{j\in\mathcal{N}} \mathrm{BCE}(\hat{o}_j, 0), \tag{5}$$

where $\hat{o}_j$ is the predicted objectness logit at cell $j$ and $\lambda_{\mathrm{noobj}} = 0.5$ down-weights the dominant background cells. The box-size and center-offset branches are regressed only at the positive cells with an $\ell_1$ loss,

$$\mathcal{L}_{\mathrm{size}} = \frac{1}{|\mathcal{P}|}\sum_{j\in\mathcal{P}} \|\hat{s}_j - s_j\|_1, \qquad \mathcal{L}_{\mathrm{off}} = \frac{1}{|\mathcal{P}|}\sum_{j\in\mathcal{P}} \|\hat{\delta}_j - \delta_j\|_1, \tag{6}$$

where $s_j$ and $\delta_j$ are the ground-truth box size and sub-pixel center offset and $\hat{s}_j$, $\hat{\delta}_j$ their predictions. The detection loss is the sum of the three terms,

$$\mathcal{L}_{\mathrm{det}} = \mathcal{L}_{\mathrm{obj}} + \mathcal{L}_{\mathrm{size}} + \mathcal{L}_{\mathrm{off}}, \tag{7}$$

and the overall training objective combines it with the attention supervision loss introduced above,

$$\mathcal{L} = \mathcal{L}_{\mathrm{det}} + \lambda_{\mathrm{att}}\,\mathcal{L}_{\mathrm{att}}, \tag{8}$$

where $\lambda_{\mathrm{att}}$ controls the relative weight of the attention supervision. All terms are optimized jointly in a single end-to-end training pass.

## IV. Result and evaluation

### A. Dataset

The experiments are conducted on a publicly available GPR dataset for intelligent recognition of subsurface utilities [46]. The dataset consists of labeled GPR B-scan images collected from real urban and industrial environments using ground-coupled antennas operating at 200 MHz and 400 MHz.

### B. Detection performance

Table I reports the detection performance under different training-data ratios. We report Precision, Recall, and F1-score. A prediction is counted as a true positive if its intersection over union (IoU) with an unmatched ground-truth box is $\geq 0.5$, following a greedy one-to-one matching in descending order of confidence; each ground-truth box can be matched at most once. Unmatched predictions and ground-truth boxes are counted as false positives and false negatives, respectively. As can be observed, the proposed method consistently outperforms the baseline across all training-data ratios. The improvement in F1-score is most pronounced at a training-data ratio of 0.7, where our method achieves 0.84 compared with 0.68 for the baseline. The training stability is also notably enhanced, with the standard deviation decreasing from 0.21 to 0.03. This gain is primarily attributed to the improvement in recall, as the attention constraint guides the network to focus on the hyperbola-band regions and thereby reduces missed detections, which demonstrates the effectiveness of the proposed method.

Nevertheless, at a training-data ratio of 0.9, no significant improvement is observed and the stability is relatively poor. This is presumably due to the limited number of samples in the test set, which leads to unstable evaluation results.

It should be noted that the above results are obtained with $\lambda_{\mathrm{att}} = 0.7$; a sensitivity analysis of the parameter $\lambda$ is further presented in the following section.

### C. Comparison of Detection Performance with Different Attention Modules

As shown in Table II, we compare the proposed method with several representative attention modules that have been applied to ground-penetrating radar, SAR, and remote-sensing recognition: Pan et al. adopt the channel attention SE, N. Wang et al. adopt the channel–spatial attention CBAM, Z. Wang et al. adopt the non-local attention Non-local, and Shen et al. adopt the coordinate attention CA. For a fair comparison, we re-implement these attention modules at the same position within an identical backbone and evaluate them on our hyperbola dataset under exactly the same training and evaluation settings, so that all results reported in the table are obtained by our own re-implementation. The proposed method achieves the best precision, recall, and F1, reaching an F1 of 0.89 with an improvement of about 9% over the second-best method. Whereas the existing modules generally struggle to balance precision and recall, our method attains the highest precision and recall simultaneously, together with the lowest F1 standard deviation, thus offering both higher detection accuracy and greater stability. This can be attributed to the fact that SE, CBAM, Non-local, and CA are all self-learned attention mechanisms that merely re-weight channel or spatial features according to the features themselves, without explicit guidance on where to attend. In contrast, our method imposes explicit supervision on the attention branch through the hyperbola-band mask, directly guiding the network to focus on the hyperbolic target regions and thereby outperforming these generic attention modules.

### D. Sensitivity analysis of the parameter $\lambda$

Fig. 5 illustrates the distribution of the detection F1 as the attention-constraint weight $\lambda$ varies from 0.1 to 10, where each value is evaluated over 10 random seeds and $\lambda = 0$ corresponds to the baseline. Across the entire range of $\lambda$, the median F1 obtained with the attention constraint consistently remains between 0.78 and 0.84, substantially higher than the baseline value of 0.68. This

TABLE I

DETECTION PERFORMANCE (MEAN ± STD OVER 10 SEEDS) AT DIFFERENT TRAINING-DATA RATIOS. "OURS" USES THE ATTENTION CONSTRAINT WITH $\lambda_{\mathrm{att}} = 0.7$. THE BEST F1 PER ROW IS SHOWN IN **BOLD**.

| Training ratio | Baseline | | | Ours | | |
|---|---|---|---|---|---|---|
| | Precision | Recall | F1-score | Precision | Recall | F1-score |
| 0.5 | $0.81 \pm 0.20$ | $0.43 \pm 0.09$ | $0.53 \pm 0.08$ | $0.86 \pm 0.08$ | $0.49 \pm 0.07$ | $\mathbf{0.62 \pm 0.06}$ |
| 0.6 | $0.71 \pm 0.21$ | $0.56 \pm 0.08$ | $0.60 \pm 0.10$ | $0.75 \pm 0.25$ | $0.65 \pm 0.06$ | $\mathbf{0.66 \pm 0.13}$ |
| 0.7 | $0.75 \pm 0.30$ | $0.68 \pm 0.10$ | $0.68 \pm 0.21$ | $0.91 \pm 0.08$ | $0.78 \pm 0.05$ | $\mathbf{0.84 \pm 0.03}$ |
| 0.8 | $0.84 \pm 0.22$ | $0.82 \pm 0.07$ | $0.81 \pm 0.16$ | $0.91 \pm 0.12$ | $0.88 \pm 0.04$ | $\mathbf{0.89 \pm 0.07}$ |
| 0.9 | $0.88 \pm 0.24$ | $0.90 \pm 0.07$ | $0.87 \pm 0.17$ | $0.85 \pm 0.22$ | $0.93 \pm 0.04$ | $\mathbf{0.87 \pm 0.17}$ |

TABLE II

COMPARISON OF DETECTION PERFORMANCE WITH DIFFERENT ATTENTION MODULES

| Method | Precision | Recall | F1-score |
|---|---|---|---|
| Pan et al. [47] | $0.86 \pm 0.20$ | $0.82 \pm 0.04$ | $0.82 \pm 0.11$ |
| N. Wang et al. [45] | $0.75 \pm 0.17$ | $0.82 \pm 0.03$ | $0.77 \pm 0.10$ |
| Z. Wang et al. [48] | $0.84 \pm 0.28$ | $0.74 \pm 0.23$ | $0.78 \pm 0.25$ |
| Shen et al. [49] | $0.80 \pm 0.23$ | $0.84 \pm 0.09$ | $0.80 \pm 0.17$ |
| **Ours** | $\mathbf{0.91 \pm 0.12}$ | $\mathbf{0.88 \pm 0.04}$ | $\mathbf{0.89 \pm 0.07}$ |

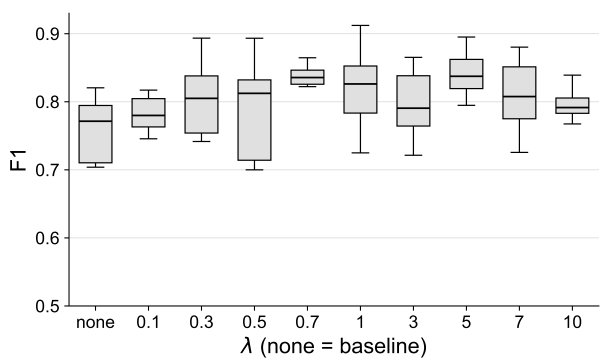



Fig. 5. lambda

indicates that the proposed method is insensitive to the choice of $\lambda$ and yields stable performance gains over a wide range of this parameter. The best result is achieved at $\lambda = 0.7$, which attains the highest median F1 of 0.84 together with the most concentrated distribution, with a standard deviation of only 0.03. When $\lambda$ is too small, the constraint becomes too weak to provide sufficient guidance, whereas an excessively large $\lambda$ slightly suppresses the primary detection task and lowers the F1 to around 0.79. These results suggest the existence of an intermediate optimal range, and $\lambda = 0.7$ is therefore adopted as the default setting throughout our experiments in consideration of both accuracy and stability.

### E. Visualization of attention maps

### F. Ablation study of NN parameters?

## V. DISCUSSION

### A. On the choice of output representation

Although our annotation tool already provides a complete parametric description of every hyperbola, we do not, at present, train the network to regress these parameters directly. There are two reasons for this. First, the mainstream detection task, together with its datasets and evaluation protocols (IoU-based precision, recall, and F1), is built around bounding boxes; retaining a bounding-box output therefore allows a fair comparison with mainstream detectors and generic attention modules under a fully identical protocol. Second, directly regressing hyperbola parameters such as curvature is sensitive to annotation noise, clutter, side lobes, and mutually overlapping echoes, while the disparate scales of the individual parameters make the training loss difficult to balance.

### B. Future work

Although directly regressing hyperbola parameters is challenging for the reasons discussed above, it remains the most informative output for practical use, and overcoming this difficulty is a natural direction for future work: moving from bounding-box localization toward predicting the parametric hyperbola itself and exporting its parameters for inversion. In GPR interpretation, inversion aims to recover the physical properties of a buried target from its hyperbolic signature: the apex position and the opening of the curve are governed by the target depth and the propagation velocity in the host medium, so that accurate parameters translate into estimates of burial depth, object radius, and the dielectric permittivity of the surrounding soil, which are the quantities of practical interest for utility mapping and condition assessment [27]. Several recent studies already regress such geometric quantities rather than boxes: GPR-Former directly predicts hyperbola parameters and refines them with a symmetry constraint [15], GPR-TransUNet regresses the curve within a transformer-augmented framework [35], and vertex-oriented detectors localize the apex directly through keypoint or state-space formulations [37], [38]. Our annotation, however, describes each target as a full hyperbolic band with vertex, width, curvature, and thickness, and a generic detection head that only regresses box size and center offset is not expressive enough to recover this richer parameterization. We therefore plan to design a dedicated prediction head and, more broadly, to fuse complementary deep-learning components, including dense detection for robust localization, band-mask segmentation for shape supervision, and transformer-style regression for the parameters, into a single model that outputs the complete parametric hyperbola end to end and thereby directly supports downstream inversion.

## VI. Conclusion

In this paper, we proposed a hyperbola-band attention method to improve hyperbola detection in GPR B-scan images. We first developed a dedicated annotation tool that describes each hyperbola as a parametric band through a few interpretable controls and exports both the shape parameters and the derived bounding box. Building on a standard bounding-box detector, we then introduced an attention branch that is explicitly supervised by the hyperbola-band mask, injecting the geometric prior into the feature representation and thereby improving detection performance.

Experiments on a public GPR dataset show that the proposed supervision consistently improves detection across training-data ratios. At a training-data ratio of 0.7, it raises the F1-score from 0.68 to 0.84 and reduces its standard deviation across random seeds from 0.21 to 0.03, with the gain driven mainly by higher recall. The method also outperforms other attention mechanisms, confirming its effectiveness. In future work, we will infer the parameters of the hyperbola itself to meet the requirements of downstream tasks such as geophysical inversion.

## Acknowledgments

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

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